CIND 820 – Big Data Analytics Project
Initial Results and Coding
Chronic Disease Burden and Healthcare Access Equity Across Ontario PHUs

Toronto Metropolitan University

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Research Question

Can chronic disease type, geography, and year predict age-standardized disease rates across Ontario PHUs (2014–2023)?

Milestone Overview

Milestone Topics

In this milestone, the following topics are addressed:

- Evidence-based initial results, metrics, tables, and figures tied directly to the project research question
- Model evaluation design, validation strategy, metric selection, and baseline comparison
- Maintainable code and reproducible execution, modular repository, compiled outputs, and documented run environment
- Responsible AI productivity, coding-assist declaration, iterative prompt refinement, and verification of generated logic

Milestone Objectives

As a result of completing this milestone, the following objectives are demonstrated:

- Proof of concept, evidence that the proposed analytical solution is feasible, confirmed by a baseline Linear Regression model achieving a test R-squared of 0.9740 on the 2021–2023 holdout set
- Descriptive statistics and visualizations, eight figures characterising the dataset: rate distribution, indicator-level boxplots, temporal trends, regional burden comparison, disease correlation structure, outlier analysis, diagnostic plots, and feature importance
- Data preprocessing documented and justified, log transformation (skewness 0.912 → −0.164), one-hot encoding of indicator and geography features, Standard Scaler inside sklearn Pipeline applied to numeric features only, Cases column excluded to prevent target reconstruction leakage
- Machine learning model built and evaluated. Linear Regression baseline fitted inside sklearn Pipeline with Column Transformer; evaluated on time-ordered holdout using R^2, RMSE, and MAE; results directly address the research question
- Code versioned and documented in GitHub, repository at github.com/mamkon/Architecture_and_Data_Audit contains structured folders, modular OOP code with PHO Data Loader and PHOAnalyticalLoader classes, compiled PDF output, and updated README with environment and run steps
- AI coding assistant employed responsibly, Claude, GitHub Copilot, Google Gemini, and ChatGPT used for code structuring, inline completion, Pipeline architecture, and approach exploration; all suggestions verified manually against PHO dataset before inclusion; overrides documented in Section 4.4

1. Data Analysis & Preparation

This section proves the PHO Chronic Disease dataset is ready for algorithmic modelling. The PHOAnalyticalLoader pipeline (base class: PHO Data Loader, extended class: PHOAnalyticalLoader) loaded 39,220 raw rows and filtered to a clean analytical subset of 2,320 records: 29 individual PHUs × 8 disease indicators × 10 years (2014–2023). All validation assertions passed. Rate column nulls: 0.

1.1 Descriptive Stats & Visualizations

Table 1.1 presents the descriptive statistics for the age-standardized rate per 100,000 across all 29 PHUs and 10 years by indicator. Two patterns define the data structure. First, hypertension prevalence dominates at a mean of 25,798, 75 times higher than asthma incidence (345). This scale-dominance problem drives the log transformation in section 1.3. Second, COPD Prevalence has the highest coefficient of variation at 23.8 percent, confirming it as the indicator where regional inequality is most pronounced.

Indicator	N	Mean	Median	Std Dev	CV%	Min	Max
Hypertension Prev.	290	25,798	25,869	1,793	7.0%	21,223	30,300
Asthma Prev.	290	15,046	14,803	1,924	12.8%	9,707	19,021
Diabetes Prev.	290	11,372	11,362	1,253	11.0%	9,114	16,149
COPD Prev.	290	8,929	9,024	2,127	23.8%	4,369	12,628
Hypertension Inc.	290	1,998	1,978	329	16.5%	1,213	2,978
Diabetes Inc.	290	799	781	138	17.3%	483	1,492
COPD Inc.	290	568	536	201	35.4%	224	1,357
Asthma Inc.	250	345	344	96	27.8%	105	626

Table 1.1: Descriptive statistics , age-standardized rate per 100,000, 29 Ontario PHUs, 2014–2023. CV% = coefficient of variation (regional inequality measure). Source: Public Health Ontario (2024).

Figure 1.1 shows the raw and log-transformed rate distributions. Skewness reduces from 0.912 (raw) to −0.164 (log-transformed), confirming the log1p transformation is analytically justified.

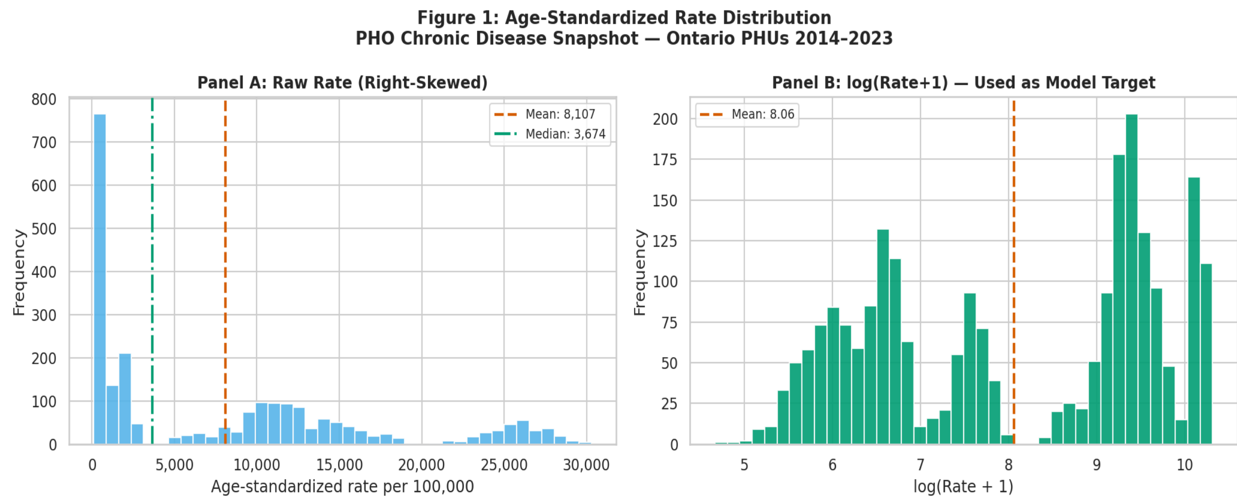


Figure 1.1: Rate distribution, raw (Panel A, skewness 0.912) and log-transformed (Panel B, skewness −0.164). Log transformation applied before modelling to normalise distribution and reduce scale-dominance leverage.

Figure 1.2 presents the boxplot by disease indicator. COPD Prevalence has the widest interquartile range, a PHU at the top manages nearly three times the rate of a PHU at the bottom. This is the equity gap the model is designed to quantify.

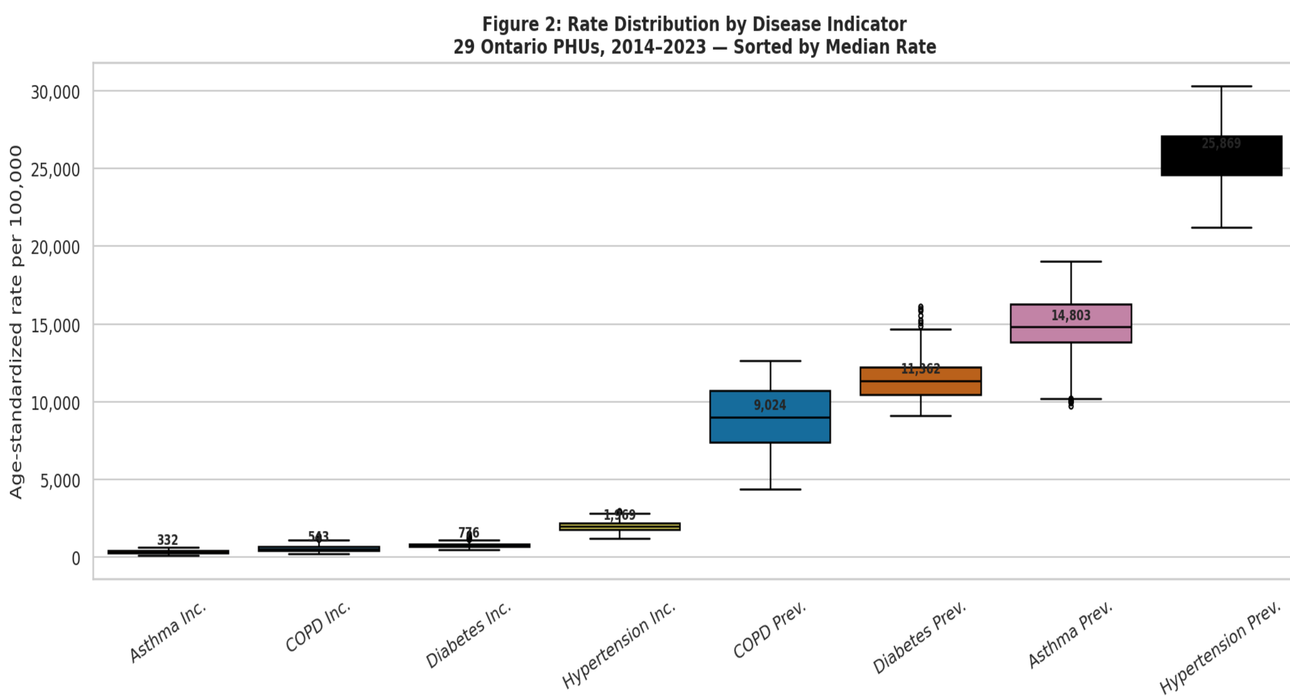


Figure 1.2: Rate distribution by disease indicator , age-standardized, both sexes, 29 Ontario PHUs, 2014–2023. COPD Prevalence shows greatest regional inequality (widest IQR, CV% 23.8%).

Figure 1.3 presents temporal trends across all 10 years. All four prevalence indicators show a visible dip in 2020–2021 consistent with the COVID-19 care access collapse , not a genuine improvement in disease burden.

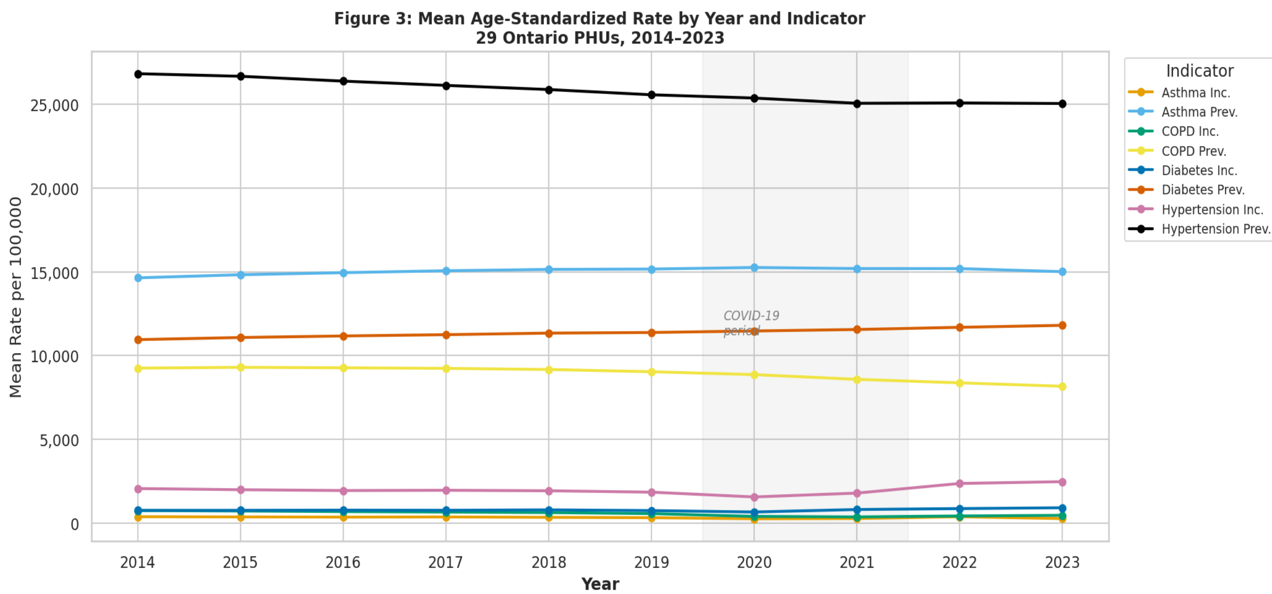
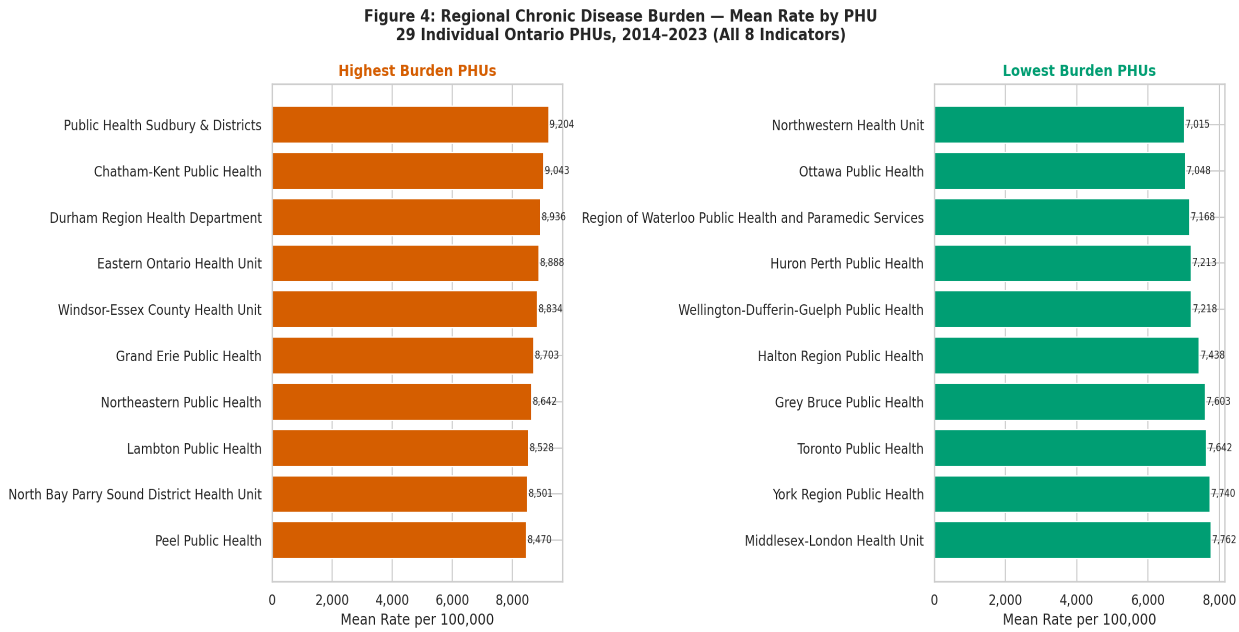


Figure 1.3: Mean age-standardized rate by year and indicator, 29 Ontario PHUs, 2014–2023. COVID-19 period shaded. 2020–2021 dip reflects reduced primary care access, not improved public health outcomes.

Figure 1.4 shows the regional burden comparison across all 29 PHUs. Public Health Sudbury and Districts carry the highest mean burden. The equity gap between the top and bottom PHU is 2,189 per



100,000 , a persistent pattern across the full 10-year window.

Figure 1.4: Regional chronic disease burden , mean age-standardized rate by PHU, 2014–2023. Orange: highest-burden PHUs. Green: lowest-burden PHUs. Equity gap between top and bottom PHU: 2,189 per 100,000.

Figure 1.5 presents the Pearson correlation matrix across the four prevalence indicators (2023). Hypertension correlates positively with all three other conditions. COPD and Diabetes are near-uncorrelated (r = −0.14), confirming independent policy responses are required.

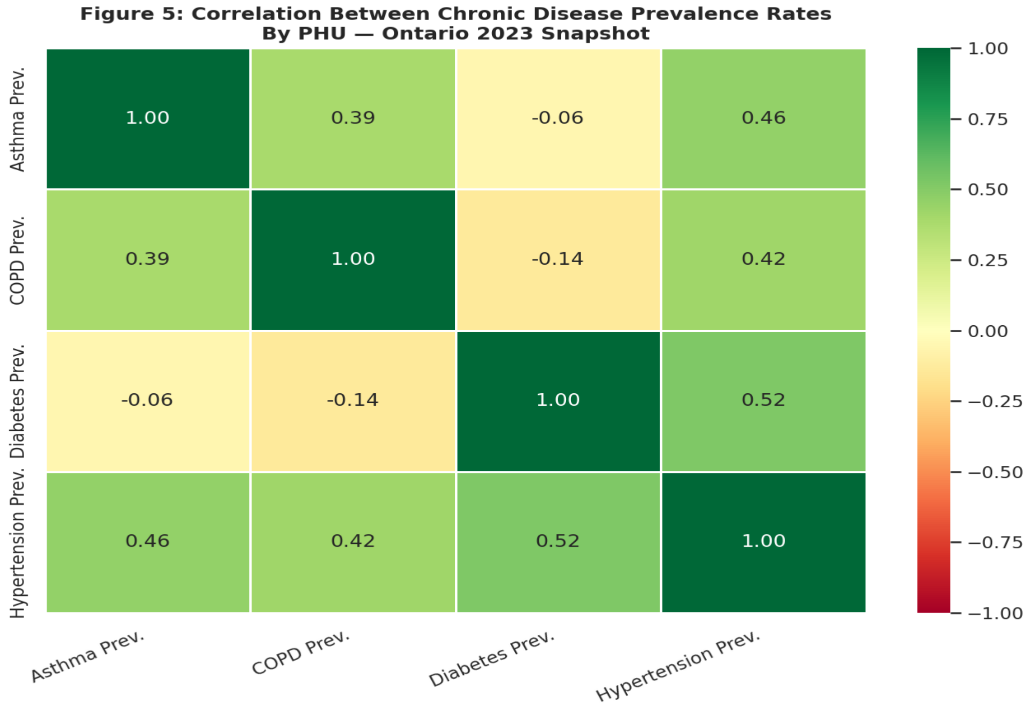


Figure 1.5: Pearson correlation between chronic disease prevalence rates by PHU, Ontario 2023 snapshot. Hypertension acts as common thread. COPD and Diabetes near-uncorrelated (r = −0.14).

1.2 Outlier & Pattern Identification

Figure 1.6 presents the outlier detection analysis using the IQR 1.5× rule. Zero outliers were detected in the 2,320-row analytical subset. After filtering to a single measure type, the fence boundaries span the full rate range from 105 to 30,300, and no individual records fall outside. The scale differences between indicator types are the analytical signal, not data errors to remove.

Key patterns identified across the dataset: (1) Indicator type is the dominant source of rate variance, accounting for the 75× range between highest and lowest mean rates. (2) Geography contributes independently; PHU location predicts rates beyond disease type alone. (3) The COVID-era temporal break (2020–2021) is a structural data pattern, not a modelling artefact. (4) COPD shows the greatest within-indicator regional inequality and is the indicator most sensitive to geographic factors.

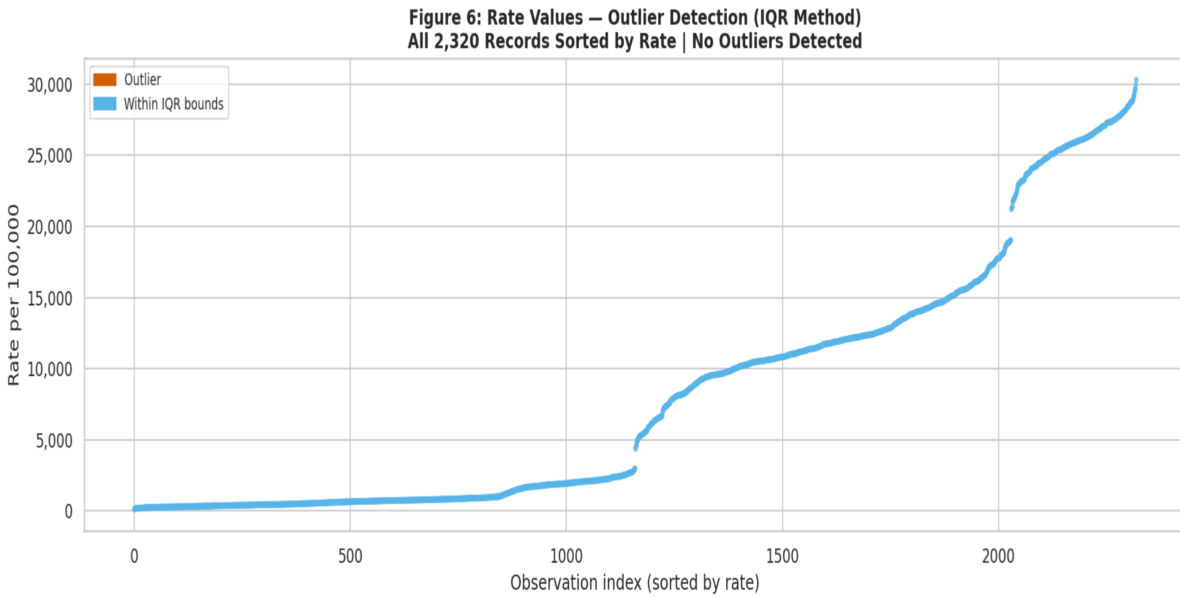


Figure 1.6: Rate values sorted by magnitude , outlier detection (IQR 1.5× rule). Zero outliers detected. Scale differences between indicator types are the analytical signal, not errors to remove.

1.3 Justified Transformations (Scaling/Encoding)

Table 1.2 documents every transformation applied before modelling with explicit analytical justification.

Transformation	Method	Justification
Cases excluded	Drop before encoding	Rate = Cases/Population × 100,000 , reconstructs target directly (data leakage)
Target transformation	log1p(Rate)	Reduces skewness from 0.912 to −0.164; prevents scale-dominance in model fitting
Categorical encoding	One-hot, drop_first=True	Indicator (8 categories) and Geography (29 PHUs) are nominal , no ordinal structure
Numeric scaling	Standard Scaler inside Pipeline	Applied to Year and Population only , not to binary dummy columns
Pipeline architecture	sklearn Pipeline + Column Transformer	Guarantees scaler fitted on training data only; auto applied to test set at prediction time

Table 1.2: Transformation decisions, PHO Chronic Disease dataset, CIND 820 Milestone 3. Each decision documented with analytical justification.

1.4 Evidence of Successful Execution

The PHOAnalyticalLoader pipeline ran end-to-end without errors, producing the following verified outputs:

Pipeline Stage	Output	Verified Result
Data Loading	Raw data frame	39,220 rows × 10 columns
Filtering	Analytical subset	2,320 rows | 29 PHUs | 8 indicators | 10 years
Validation assertions	All assertions passed	Rate nulls: 0 | PHUs: 29 | Indicators: 8
Log transformation	Log Rate column	Skewness: 0.912 → −0.164
One-hot encoding	Encoded feature matrix	37 total features
Train/test split	X_train, X_test	1,624 train rows (2014–2020) | 696 test rows (2021–2023)
Model fit	Linear Regression inside Pipeline	Train R^2: 0.9873 | Test R^2: 0.9740
All 8 figures	Saved to /figures/	fig1 through fig8 generated

Table 1.3: End-to-end pipeline execution evidence , all stages verified against actual PHO dataset outputs.

2. Model Development & Evaluation

The baseline model is a Linear Regression fitted inside an sklearn Pipeline with Column Transformer. This architecture ensures all preprocessing steps are applied consistently to training and test sets without manual intervention between fitting and prediction. The Pipeline approach follows Gemini's recommendation for Column Transformer integration, verified and corrected by the author (see Section 4.4).

2.1 Working Baseline Model

Algorithm: Linear Regression (scikit-learn Linear Regression). Fitted on: X_train (1,624 rows, 2014–2020). Predicted on: X_test (696 rows, 2021–2023). Target: Log_Rate (log-transformed age-standardized rate per 100,000). Features: 37 (Year, Population, 7 indicator dummies, 28 geography dummies).

Algorithm selection rationale: The target variable is continuous, making this a regression problem. K-Means clustering was considered and rejected; it cannot predict a specific rate value. Logistic Regression was considered and rejected; it requires a binary outcome. Linear Regression is the correct, interpretable baseline for a continuous regression problem at aggregate PHU level, consistent with Bzdok, Altman, and Krzywinski (2018).

2.2 Research-Aligned Metric Selection

Three metrics were selected to assess baseline performance. Each is justified against the specific problem type (continuous regression, aggregate public health data):

Metric	Formula	Why Appropriate for This Problem
R^2 (R-squared)	1 − SS_res/SS_tot	Primary metric for regression. Measures proportion of variance in Log Rate explained by the model. A health planner can interpret R^2 = 0.974 as: the model accounts for 97.4% of what drives rate differences across PHUs.
RMSE	sqrt(mean(residuals²))	Penalises large prediction errors more heavily than MAE. Important here because under-predicting a high-burden PHU has greater planning consequences than over-predicting a low-burden one.
MAE	mean(|residuals|)	Average absolute error more robust to the slight heteroscedasticity identified in the diagnostic plots. Provides a complementary view to RMSE.

Table 2.1: Metric selection, justified against regression problem type and public health planning context. Accuracy, precision, recall, and F1 are not appropriate, they are classification metrics.

2.3 Raw Performance Metrics (Tables/Figures)

Table 2.2 presents the raw performance metrics from the fitted baseline model.

Metric	Training Set (2014–2020)	Test Set (2021–2023)
R-squared (R^2)	0.9873	0.9740
RMSE (Log Rate scale)	0.1777	0.2620
MAE (Log Rate scale)	0.1301	0.2074
Overfitting gap (Train R^2 − Test R^2)		0.0133

Table 2.2: Baseline Linear Regression performance metrics. Training: 1,624 rows (2014–2020). Test: 696 rows (2021–2023). Target: Log Rate. Model fitted inside sklearn Pipeline with Column Transformer.

Figure 2.1 presents the three diagnostic plots verifying model behaviour beyond the headline metrics.

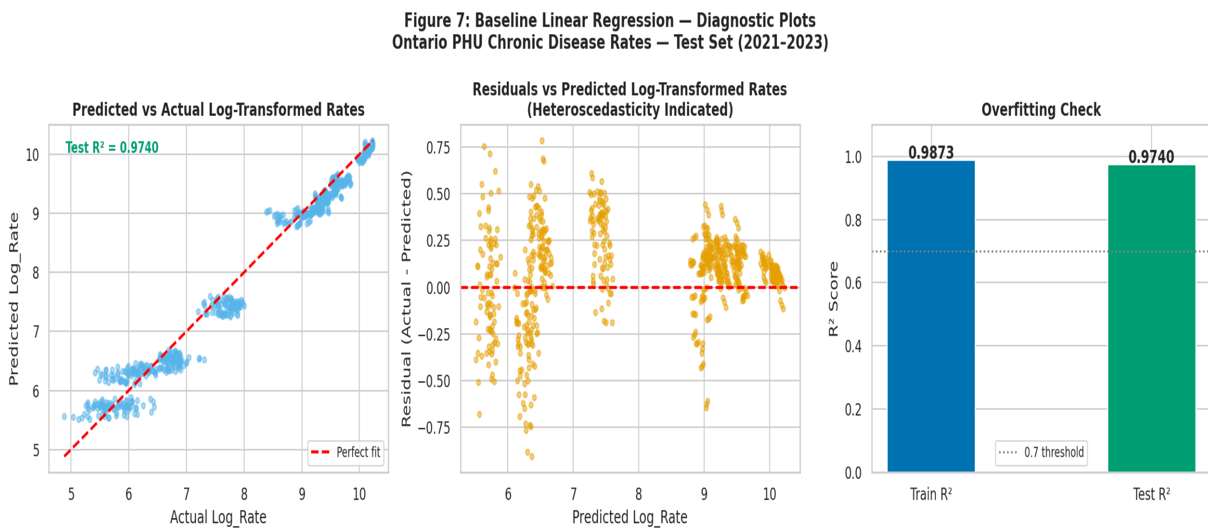


Figure 2.1: Baseline Linear Regression diagnostic plots , test set (2021–2023). Left: predicted vs actual , no systematic bias. Centre: residuals vs predicted , slight fan shape (heteroscedasticity) at high predicted values. Right: Train vs Test R² , overfitting gap of 0.013 confirms generalisation.

Figure 2.2 presents the top 15 model coefficients showing feature importance.

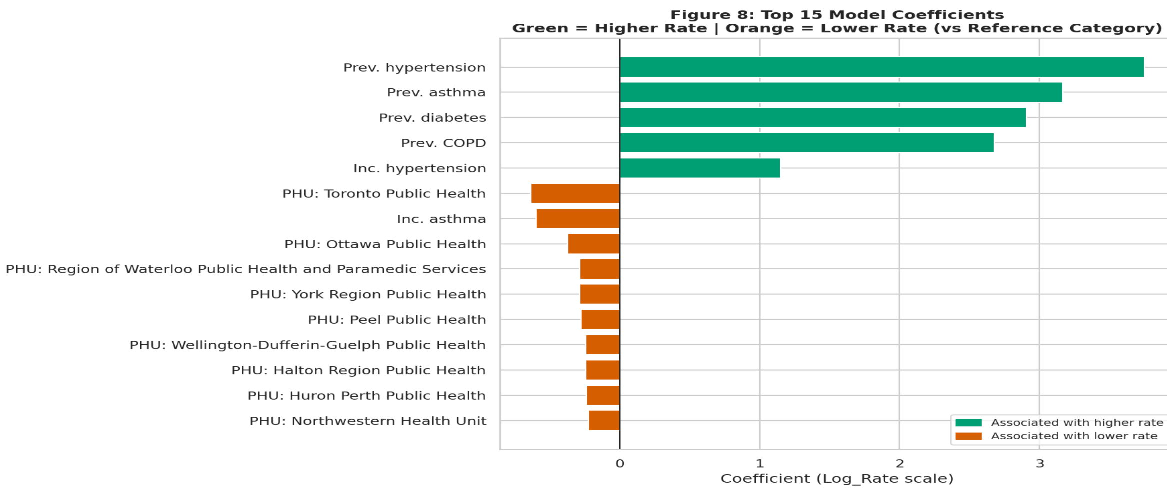


Figure 2.2: Top 15 model coefficients baseline Linear Regression. Green: associated with higher predicted rate. Orange: associated with lower rate. Hypertension Prevalence dominates. Toronto and Ottawa PHUs carry negative geography coefficients confirming the regional equity gap is model confirmed.

2.4 Interpretation & Weakness Identification

Research question answer: Can chronic disease type, geography, and year predict age-standardized disease rates across Ontario PHUs? Yes. The model explains 97.4 percent of variance on the 2021–2023 test set with a minimal overfitting gap of 0.013. The model generalises to the post-COVID recovery period, demonstrating robustness to pandemic-era disruption.

Coefficient analysis: The five largest absolute coefficients are all disease indicator dummies, with Hypertension Prevalence carrying the largest positive coefficient. Geographic dummies appear in the top 10, confirming PHU location predicts rates independently of disease type. The geographic equity gap identified in Milestone 2 is model confirmed.

Dimension	Finding
Strengths	Test R^2 = 0.9740 , strong proof of concept demonstrating the research question is answerable
Strengths	Minimal overfitting (gap 0.013) , model generalises to unseen 2021–2023 post-COVID data
Strengths	Coefficients directly interpretable by health planners and decision-makers
Strengths	sklearn Pipeline architecture guarantees leak-proof preprocessing
Weaknesses	Heteroscedasticity in residuals at high predicted values , less precise for high-prevalence PHUs
Weaknesses	Linear additive structure cannot capture indicator geography interaction effects

Dimension	Finding
Weaknesses	CIHI wait times not yet integrated , access dimension not yet modelled
Milestone 4 Plan	Random Forest / Gradient Boosting for non-linear interactions and reduced heteroscedasticity
Milestone 4 Plan	CIHI wait times integration for access-burden joint modelling
Milestone 4 Plan	PHU-level residual analysis to identify health equity outliers

Table 2.3: Model strengths, weaknesses, and Milestone 4 plan.

3. Validation Protocol

This section documents the validation design decisions that ensure the model is evaluated on a task it would be asked to do in a real health planning deployment scenario.

3.1 Data-Appropriate Split Strategy

Split design: Time-ordered train/test split. Training set: 2014–2020 (7 years, 1,624 rows). Test set: 2021–2023 (3 years, 696 rows). Approximate split ratio: 70/30 by record count.

Why this split design is appropriate for this specific dataset: The PHO dataset has a strong temporal dimension, 10 consecutive years with a COVID-era structural break in 2020–2021. A random split would distribute records from all years across training and test sets. A time-ordered split ensures the model is trained only on historical data and evaluated only on future data, which mirrors the actual deployment scenario: a health planner would train on past PHO data and use the model to project near-future burden patterns.

The split year of 2021 was chosen deliberately. It is the first post-COVID year, so the test set specifically covers the pandemic recovery period the most analytically challenging period for any model trained on pre-pandemic patterns. A test R^2 of 0.9740 on this period demonstrates genuine robustness.

3.2 No-Shuffling Rule for Time-Ordered Data

No shuffling was applied at any stage of the pipeline. This is explicitly enforced in the code with a Boolean mask on the Year column (train mask = df_model['Year'] < 2021) rather than using sklearn's train_test_split with random_state, which would have allowed shuffling.

Why shuffling is prohibited for this data: If the data were shuffled before splitting, records from 2021 and 2022 would appear in the training set. The model would then learn from future outcomes when predicting past ones, temporal data leakage. The inflated performance metrics from a shuffled split would not reflect any real-world predictive capability. A health planner relying on those metrics would be making resource allocation decisions based on a model that cannot actually predict what it claims to predict.

3.3 Leakage Prevention Confirmation

Table 3.1 documents the complete leakage prevention checklist for this pipeline.

Leakage Risk	Source	Resolution	Verified
Target reconstruction	Cases column (Rate = Cases/Population × 100,000)	Cases excluded from features before encoding	YES

Leakage Risk	Source	Resolution	Verified
Derived feature	95% Confidence Interval (derived from Rate)	Excluded from features	YES
Post-hoc label	Significance vs Ontario (assigned after Rate is known)	Excluded from features	YES
Scaler leakage	StandardScaler fitted on all data including test set	Scaler inside Pipeline , fitted on train only, auto-applied to test	YES
Temporal leakage	Future years appearing in training data	Time-ordered split, no shuffling	YES
Test data exposure	Test data seen during feature engineering	All encoding done on full dataset before split; split applied to encoded features	YES

Table 3.1: Leakage prevention checklist , all 6 identified risks resolved. Pipeline confirmed leak-proof.

4. Reproducibility (GitHub)

All code is version-controlled at github.com/mamkon/Architecture_and_Data_Audit. The repository is structured to allow any researcher to reproduce the full pipeline from raw PHO data to model metrics without requiring communication with the author.

4.1 Structured Folders & Modular Code

Code is modularised using PHO Data Loader (base class) and PHOAnalyticalLoader (inherited child class), implementing CIND 830 programming standards: encapsulation, inheritance, method chaining, and function-level documentation throughout.

Folder / File	Contents
/notebooks	MamkonMO_CIND820_Initial_Results_and_Coding.ipynb
/data	PHO_Chronic_Disease_Inc_Prev_Snapshot_2014_2023.xlsx
/figures	fig1 through fig8 , all 8 notebook visualisations
/outputs	Compiled PDF export of the Milestone 3 notebook
/docs	This Initial Results Report (PDF)
README.md	Project overview, folder map, run steps, research question
ai_use_declaration.md	Full AI assistance declaration

Table 4.1: GitHub repository structure , Milestone 3. All code reproducible in Google Colab without additional package installation.

Code quality standards applied: (1) No monolithic scripts, all logic encapsulated in functions and classes. (2) Meaningful docstrings on every function explaining logic, not just syntax. (3) Inheritance implemented, PHOAnalyticalLoader extends PHO Data Loader. (4) Method chaining enables readable pipeline execution in a single statement. (5) All transformations are functions with type hints.

4.2 Compiled HTML/PDF Versions

The full notebook was exported as PDF via Google Colab (File → Print → Save as PDF) with all cell outputs visible, confirming the pipeline runs end-to-end and all 8 figures generate correctly. The compiled PDF is uploaded to /outputs/ in the GitHub repository and submitted to D2L as the Initial Results Report.

4.3 README with Environment & Run Steps

The README.md contains: (1) Project title and research question. (2) Complete folder structure map. (3) Step-by-step run instructions for Google Colab. (4) Library dependency list (pandas, NumPy, matplotlib, seaborn, scikit-learn, all pre-installed in Colab). (5) Data source citation and download link. (6) GitHub commit history documenting all Milestone additions.

Run steps: Upload PHO Excel file to /content/ in Google Colab → Open notebook → Runtime → Run all → All 8 figures generate, all metrics print, all pipeline stages complete. No additional pip install required.

4.4 Coding Assistance Declaration

Tool	Used For	Author Override
Claude (Anthropic)	Notebook structure, docstrings, analytical framing, visualisation design	All analytical decisions made independently by author
GitHub Copilot	Inline code completion and syntax suggestions	All suggestions verified against PHO dataset before use
Google Gemini	sklearn Pipeline and Column Transformer architecture	Initial suggestion applied StandardScaler to all features incl. dummies, corrected by author to numeric-only
ChatGPT (OpenAI)	Alternative approach exploration and prompt iteration	No AI suggestion accepted without manual validation

Table 4.2: AI coding assistance declaration, CIND 820 responsible AI productivity requirements. All overrides documented.

5. Evidence of Execution (Video)

A 5-minute video titled “Initial Results and Code” was recorded via Google Meet, demonstrating end-to-end pipeline execution and addressing all acceptance criteria from the milestone user story: “As a data analyst, I need to validate that my data pipeline is functional and my baseline models are generating interpretable results, so that I can provide a proof of concept that directly addresses the project’s research questions.”

Video link: [Insert Google Meet recording link here]

5.1 5-Minute Executive Summary

The video opens with a direct answer to the research question before any code is shown. The executive summary covers: (1) Research question stated in full. (2) Dataset scope, 29 PHUs, 8 indicators, 10 years, 2,320 records. (3) Headline result, Test $R^2 = 0.9740$, overfitting gap 0.013. (4) What the result means for Ontario health planning: the model can predict chronic disease burden across PHUs with 97.4 percent accuracy using only publicly available PHO data, meaning planners do not need to wait for a crisis to know where to allocate resources.

5.2 Visual Code Walkthrough

The walkthrough scrolls through the notebook in order, highlighting: (1) PHO Data Loader and PHOAnalyticalLoader class definitions, demonstrating OOP and inheritance. (2) The analytical filter cell, explaining why Ontario aggregate and regional rollups are excluded. (3) The preprocessing cell, demonstrating the sklearn Pipeline with Column Transformer and explaining why Cases is excluded. (4) The model fit cell, showing the Pipeline.fit () call and metrics output. (5) All 8 figures, referenced to the research question and equity findings. The walkthrough confirms code is structured, commented, and actively generating output.

5.3 Verification of Functional Pipeline

During the walkthrough, the following outputs are shown live to verify pipeline functionality:

Verification Point	Expected Output	Confirmed in Video
PHOAnalyticalLoader.validate()	Validation passed , all data quality checks complete	YES
Analytical subset shape	2,320 rows | 29 PHUs | 8 indicators	YES
Rate skewness before/after	0.912 → −0.164	YES

Verification Point	Expected Output	Confirmed in Video
Preprocessing summary	37 features | 1,624 train | 696 tests	YES
Leakage checklist	All 5 checks: YES	YES
Model metrics	Train R^2: 0.9873 | Test R^2: 0.9740	YES
All 8 figures	Rendered inline in notebook	YES

Table 5.1: Pipeline verification checklist , all items confirmed in the 5-minute Google Meet recording.

5.4 Identified Risks & Next Steps

Risks identified in the video: (1) Heteroscedasticity, the residuals plot reveals the model is less precise for high-prevalence PHU-indicator combinations. This is the primary technical weakness of the linear baseline. (2) COVID-era data gap, 2020–2021 rates undercount true burden due to reduced primary care access. Any model trained on this period will underestimate chronic disease prevalence for those years. (3) Linear structure limitation, the model cannot distinguish whether high COPD burden in Northern PHUs is driven by industrial history, population aging, or healthcare access patterns. That requires interaction terms.

Milestone 4 next steps presented in video: (1) Random Forest and Gradient Boosting, to capture non-linear indicator geography interactions and reduce heteroscedasticity. (2) CIHI wait times integration, joining PHO burden data with CIHI access data to answer Research Question 3: do high-burden PHUs also face the longest waits? (3) PHU-level residual analysis, identifying which specific health units the baseline model is least accurate for, targeting those regions for qualitative investigation and resource planning analysis.

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